

Fabric Wrap a Gift Order SOP (1x Episode: 1 Order)

One episode wraps **one** boxed item. The box is carried from the order spot onto a cloth square in the working area, the near and far corners of the cloth are folded over the box, the left and right corners are knotted on the top face, one gift card is tucked under the knot, one band is put around the body, and the finished bundle is staged in the outbound lane.

The order never changes: lay the cloth, set the box on it, fold the near corner over, fold the far corner over, knot the two side corners, tuck the card, band the body, then stage the bundle. Nothing is knotted before both flaps are over the box, the card goes in before the band, and the band never goes on before the knot is tight.

The left gripper owns the cloth and the box. It lays the cloth out, carries the box (in Config L and M), pins the folded flaps down, brings the left corner up, and steadies the bundle. The right gripper does the working motions. It folds both flaps over, brings the right corner up, ties the knot, tucks the card, lifts the band off the band post, and carries the bundle to the outbound lane (in Config L and M). In Config R the box and the lane swap sides, so the right gripper carries the box and the left gripper carries the bundle. The knot and the band are the two places where both grippers work together, and each bullet says what each one does.

Exactly one box, one cloth square, one gift card, and one band are handled per episode. The cloth always goes down pattern side down, so the finished bundle shows the pattern on every face.

The table is set up in one of three ways. Only the box moves, and in Config R the outbound lane swaps to the back left to make room for it; the cloth stack, the card stack, the band post, and the working area are in the same place in all three.

- **Config L:** the box is on the order spot at the back left.
- **Config M:** the box is on the order spot at the front center, in front of the working area.
- **Config R:** the box is on the order spot at the back right, and the outbound lane is at the back left.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the box starts at the back left (**Config L**), the front center (**Config M**), or the back right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.

- **Same-side rule:** the gripper on the box's side carries it onto the cloth — the left gripper in Config L and M, the right gripper in Config R — and the gripper on the outbound lane's side carries the bundle to it — the right gripper in Config L and M, the left gripper in Config R. No arm reaches across the table.
- **Fixed roles:** the cloth, the flaps, the knot, the card, and the band are handled the same way in all three configs, and the working area is always the center of the table.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the order spot in the start zone for this episode's config, the cloth stack at the front left, the working area at the center, the band post at the right edge, the card stack at the front right, and the outbound lane at the back right (back left in Config R).
3. The working area is visible from above, so the four cloth corners, the knot on the top face, and the band can all be seen.
4. Both arms are at home with grippers open.
5. The table is clean, dry, and bare apart from the boxes, the cloth stack, the card stack, the band post, and the outbound lane.
6. The left arm reaches the cloth stack, the working area, (Config L and M) the order spot, and (Config R) the outbound lane at the back left. The right arm reaches the working area, the band post, the card stack, (Config L and M) the outbound lane at the back right, and (Config R) the order spot at the back right.

Materials checklist

1. One or more **boxed items** sit on the order spot for this episode's config, and the other two start zones are bare. Each one is closed, sound, and light enough for one gripper: no crushed corner, no open flap, no loose tape.
 - **Config L:** back left
 - **Config M:** front center, in front of the working area, clear of the cloth stack and the card stack
 - **Config R:** back right

2. A **cloth stack** of square cloths lies at the front left, folded, pattern side up on the stack. Each cloth is clean, dry, and not frayed at the corners. A cloth is the right size when, with the box in the middle, the near and far corners reach past the top face and the left and right corners still have enough length to knot on the top face.
3. A **card stack** of gift cards sits at the front right, writing side up, cards flat and uncreased.
4. A **band post** stands at the right edge with elastic **bands** hanging over it. Each band is sound and stretches to open wider than a wrapped bundle without going slack.
5. The working area at the center is clear.
6. The **outbound lane** at the back right (Config L and M) or the back left (Config R) is clear, or holds finished bundles from earlier episodes in one row.

Workspace layout

- **Order spot:** the boxed items waiting to be wrapped — back left (**Config L**), front center, in front of the working area (**Config M**), or back right (**Config R**). One per episode.
- **Cloth stack:** front left, the folded cloth squares.
- **Working area:** center, where the wrapping, the knot, the card, and the band happen.
- **Band post:** right edge, the bands hanging on their post.
- **Card stack:** front right, the gift cards.
- **Outbound lane:** back right in Config L and M, back left in Config R, the finished bundles in one row.

Everything the left arm handles is staged on the left. Everything the right arm handles is staged on the right. Neither arm reaches across to the other side for a supply, the box, or the lane.

Arm assignments

- **Left gripper:** lays the cloth square out flat, carries the box from the order spot onto the cloth in Config L and M, pins the two folded flaps to the top face, brings the **left corner** up and holds it at the middle of the top face, makes the second pass of the knot, holds the bundle steady for the card tuck, takes the low side of the band, and in Config R carries the finished bundle to the outbound lane at the back left.
- **Right gripper:** in Config R, carries the box from the order spot at the back right onto the cloth; folds the near corner and the far corner over the box, brings the **right corner** up, makes the first pass of the knot, tucks the gift card, lifts the band off the band post, and in Config L and M carries the finished bundle to the outbound lane at the back right.

Vocabulary

- **Order:** one boxed item, wrapped whole in a single episode.
- **Order spot:** where the box waits at the start of the episode — back left (**Config L**), front center, in front of the working area (**Config M**), or back right (**Config R**). One per episode,

chosen before recording and never changed mid-episode.

- **Box:** the closed box holding the order. Its **top face** points up the whole time, and its **body** is the four faces around the sides.
- **Cloth square:** one square cloth. Its **pattern side** is the printed face and ends up facing out, so the cloth is laid pattern side down.
- **Diamond:** how the cloth is laid out — turned so one corner points toward the front edge of the table, one points away from it, and one points to each side. The cloth is a diamond and the box on it stays square to the table.
- **Near corner:** the cloth corner toward the front edge. **Far corner:** the one away from it. **Left corner** and **right corner:** the two side corners.
- **Flat:** the cloth shows no ridge, fold line, or bunched patch when you look across it from the side.
- **Flap:** the near corner or the far corner once it lies folded over on the top face.
- **Gathered rope:** a side corner squeezed along its length into a rope-like bundle of cloth, so it ties into a small tight knot instead of a wide flat one.
- **Half knot:** one pass of one gathered rope over the other and back up under it, pulled tight.
- **Square knot:** two half knots, with the second pass made the opposite way round to the first, so the knot lies flat on the top face and holds. Two passes the same way round make a knot that stands on end and slips loose — that is not a square knot.
- **Tail:** one of the two lengths of cloth standing up out of the knot.
- **Bundle:** the box once the cloth is knotted around it.
- **Gift card:** one card tucked under the knot so it stands up between the two tails, writing side toward the front edge.
- **Band:** one elastic band, put around the middle of the body to hold the wrap closed.
- **Band post:** the upright peg at the right edge that the bands hang on.
- **Outbound lane:** the printed strip where finished bundles stand in one row — at the back right in Config L and M, at the back left in Config R.
- **Clear gap:** two bundles in the lane are close but not touching, with table showing between them.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Release point:** over the working area, over the band post, over the card stack, or over the outbound lane. A gripper opens nowhere else.

Steps

Step 1 lays the cloth and sets the box on it. Steps 2 and 3 close the cloth around the box and knot it. Steps 4 and 5 add the card and the band. Step 6 stages the bundle and Step 7 ends the episode.

The box pick in Step 1 and the carry in Step 6 depend on the config: in Config L and M the **left gripper** takes the box from the order spot and the **right gripper** carries the bundle to the lane at the back right; in Config R the **right gripper** takes the box from the back right and the **left gripper** carries the bundle to the lane at the back left. Every other line is the same in all three configs.

Step 1: Lay the cloth and set the box on it

Goal: one box sits in the middle of a flat cloth square, top face up, with the cloth laid as a diamond.

- The **left gripper** closes on the top **cloth square** of the cloth stack at the front left, lifts it clear, carries it to the working area, and lays it flat at the center **pattern side down**.
- The **left gripper** turns the cloth into a **diamond**, so one corner points toward the front edge, one points away from it, and one points to each side.
- The **left gripper** runs one flat sweep from the middle of the cloth out to each corner in turn, until the cloth is **flat**. Look where the box is before reaching for it.
- **IF the box is at the back left (Config L):** the **left gripper** closes on one **box** on the order spot, lifts it clear of the table, carries it forward and to the right to the working area, and sets it down in the middle of the cloth **top face up**, squared to the table so its edges run front to back and left to right.
- **IF the box is at the front center (Config M):** the **left gripper** closes on one **box** on the order spot, lifts it clear of the table, carries it straight back to the working area, and sets it down in the middle of the cloth **top face up**, squared to the table the same way.
- **IF the box is at the back right (Config R):** the **right gripper** closes on one **box** on the order spot, lifts it clear of the table, carries it forward and to the left to the working area, and sets it down in the middle of the cloth **top face up**, squared to the table the same way.

Then, in all three:

- The gripper that set the box down releases and lifts clear.

Take one cloth and one box only. Never drag or slide a box across the table, and never lay the cloth pattern side up.

Check: the cloth lies flat as a diamond, the box sits square in the middle of it, and all four cloth corners stick out past the box by about the same amount. If one corner is short, the **left gripper** slides the box until the four corners look even again.

Expected state: one box rests top face up on a flat cloth diamond, with cloth showing all the way round it.

Step 2: Fold the near and far corners over the box

Goal: the box is covered front and back by two flaps lying flat on the top face, held down.

2.1 Bring the near corner over

- The **right gripper** pinches the **near corner** of the cloth at its tip and lifts it up and over the box.
- Lay the corner flat on the **top face** and press it down.
- If the tip hangs past the far edge of the box, fold it back on itself so it lies on the top face.
- The **left gripper** presses down on the corner near the far edge of the top face and holds it there.

2.2 Bring the far corner over

- The **right gripper** pinches the **far corner** at its tip and lifts it up and over the box, toward the front edge.
- Fold the tip under along its leading edge so no raw point shows, then lay the corner flat on top of the near corner and press it down.
- The **left gripper** keeps holding both flaps down against the top face.

The far corner always goes on top of the near corner. If a flap springs back, if a box face still shows front or back, or if a flap is bunched instead of flat, the **right gripper** lifts that corner and lays it again.

Check: no box face shows at the front or the back, both flaps lie flat on the top face with the far corner on top, and the **left gripper** is still holding them down.

Step 3: Knot the two side corners

Goal: the two side corners are tied into a **square knot** sitting on the middle of the top face, with the cloth tight on every face.

The **left gripper** does not let go of the flaps until its own corner is holding them down in 3.1.

3.1 Bring the left corner up

- The **right gripper** presses down on the middle of the top face, holding both flaps.
- The **left gripper** lets go, pinches the **left corner** at its tip, and slides down the corner toward the box until the cloth squeezes into a **gathered rope**.
- The **left gripper** lifts the gathered rope up and over onto the middle of the top face and holds it down there.
- The **right gripper** lets go.

3.2 Bring the right corner up

- The **right gripper** pinches the **right corner** at its tip and gathers it into a **gathered rope** the same way.

- The **right gripper** lifts it up and over and lays it across the left gripper's rope at the middle of the top face.

3.3 Tie the square knot

- The **right gripper** lays its rope over the left gripper's rope, passes it under, and brings it back up.
- Both grippers pull their own rope apart until this **half knot** seats down on the middle of the top face and the cloth pulls tight on every face.
- The **left gripper** now lays its rope over the right gripper's rope, passes it under, and brings it back up — the opposite way round to the first pass.
- Both grippers pull their own rope apart until the **square knot** is tight and sits on the middle of the top face.

Pull the cloth tight by pulling the two ropes apart, never by dragging the bundle across the table. If the knot slips, stands on end, lands off the middle of the top face, or leaves a flap loose, both grippers untie it and tie it again.

Check: one square knot sits on the middle of the top face, the cloth is tight with no loose flap and no box face showing, and the two **tails** stand up out of the knot at about the same length.

Step 4: Tuck the gift card

Goal: one gift card stands under the knot, writing side toward the front edge.

- The **left gripper** closes on the **bundle** at the middle of its left side and holds it steady.
- The **right gripper** picks one **gift card** off the card stack at the front right, **writing side** toward the front edge.
- Slide the bottom edge of the card under the near side of the knot until the card stands up between the two **tails**, then release.

One card per episode. Never lay a card loose on the bundle, never push a card under a flap or under the cloth, and never force it in far enough to loosen the knot.

Check: exactly one card stands under the knot, writing side toward the front edge, it stays put when the **right gripper** opens, and the knot is still tight.

Step 5: Band the bundle

Goal: one band lies flat around the middle of the body, holding the wrap closed.

- The **right gripper** pinches the top of one **band** on the band post at the right edge and lifts it off.
- The **left gripper** closes on the low side of the band loop.
- Both grippers pull apart until the band is open wider than the bundle.

- Both grippers lower the open band down over the knot, the tails, and the card without touching them, until the band sits around the middle of the **body**.
- Both grippers open at the same time, so the band closes onto the cloth.

One band per episode. Never roll a band up the outside of the bundle from the table, and never let a band go before it reaches the middle of the body.

Check: one band lies flat around the middle of the body, not twisted and not slack, the knot and both tails are clear above it, the card still stands, and the cloth is still tight.

Retry: if the band lands on the knot, catches the card, comes out twisted, or hangs loose, both grippers take it back off, open it wide again, and lower it back on.

Step 6: Stage the bundle to the outbound lane

Goal: the finished bundle stands in the outbound lane.

- **IF Config L or M:** the **right gripper** closes on the **body** at the middle of its right side, below the band and clear of the knot, the tails, and the card. **IF Config R:** the **left gripper** closes on the **body** at the middle of its left side, below the band and clear of the knot, the tails, and the card.
- Lift the bundle clear of the table and carry it back to its own side into the **outbound lane**.
- Lower it until it sits flat, knot up and card facing the front edge, squared to the lane, then release.
- If a bundle is already in the lane, set the new one beside it with a **clear gap**, lined up front to back and not touching.

Grip the body only. Never close on the knot, a tail, the card, or the band; never set a bundle on top of another bundle; and never drag a finished bundle across the table.

Check: the bundle stands knot up in the outbound lane in one row with any earlier bundles, the knot is still tight, the card still stands, and the band is still flat around the body.

Step 7: Confirm the order and end the episode

- Confirm the cloth covers the box with no box face showing, one square knot sits on the middle of the top face with both tails up, one card stands under the knot writing side toward the front edge, one band lies flat around the middle of the body, the bundle is in the outbound lane, the cloth stack, the card stack, and the band post are in place, and the working area is clear.
- Return both arms home, clear of the working area and the outbound lane, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

Between episodes of a batch

1. Leave the finished bundle in the outbound lane and the remaining boxes on the order spot.
2. Restack the cloth stack at the front left so the top cloth is flat, folded, and pattern side up.
3. Clear the working area of loose cloths, cards, and bands.
4. Confirm the card stack is at the front right writing side up and the band post at the right edge still carries bands.
5. Run the Setup checklists again before the next episode.

At the end of a batch

1. Carry the bundles from the outbound lane to the working area.
2. Take the band off, take the card out, untie the knot, and unfold the cloth from each box.
3. Fold each cloth square and set it back on the cloth stack at the front left, pattern side up. Replace a cloth that is stained, torn, or frayed at a corner.
4. Put the bands back on the band post and the cards back on the card stack, and throw away any band that has stretched slack or any card that is bent or torn.
5. Set the bare boxes back on the order spot for the next episode's config — back left (Config L), front center (Config M), or back right (Config R) — with the outbound lane on the back corner the config calls for, so the input matches the setup state.
6. Wipe the table and confirm the surface is clean, dry, and bare apart from the boxes and the supplies in their zones.
7. Replace a box that is crushed, split, or will not stay closed.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config L (box on the order spot at the back left, outbound lane at the back right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the box is not on the order spot for the config; a gripper reaches across the table for the box; in Config R the bundle is carried to the back right, or the right gripper carries it to the back left; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper takes the box in Config L and M and the right gripper in Config R; the right gripper carries the bundle to the lane at the back right in Config L and M and the left gripper to the lane at the back left in Config R; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the box is before the first reach, then follow that config's IF lines through Steps 1 and 6.

Violation: Wrong pickup

- **Visible cue:** the box is taken from the working area, the right side, or the outbound lane instead of the order spot, more than one box is lifted in one grasp, more than one cloth is pulled off the cloth stack, or the box is dragged or slid across the table instead of lifted and carried.
- **SOP rule broken:** Step 1, the left gripper lifts one cloth square off the cloth stack and one box off the order spot and carries each of them to the working area.
- **Coaching note:** one cloth and one box per episode, lifted clear of the table and carried, never dragged.

Violation: Cloth laid wrong

- **Visible cue:** the cloth is laid pattern side up, it is left square to the table instead of turned into a diamond, or the flat sweep is skipped and the cloth still shows fold lines, ridges, or a bunched patch when the box goes on it.
- **SOP rule broken:** Step 1, the left gripper lays the cloth flat at the center pattern side down as a diamond and sweeps it flat from the middle out to each corner.
- **Coaching note:** pattern against the table, corners pointing front, back, left, and right, and swept flat before the box lands. Everything after this depends on it.

Violation: Box set wrong on the cloth

- **Visible cue:** the box lands off the middle of the cloth so the four corners stick out by clearly different amounts, it sits turned at an angle instead of squared to the table, it is set down top face down, or it lands off the cloth altogether and the wrap goes on anyway.
- **SOP rule broken:** Step 1, the left gripper sets the box in the middle of the cloth top face up and squared to the table, with all four corners sticking out by about the same amount.
- **Coaching note:** center it and square it before letting go. A box set off center runs out of cloth at the knot.

Violation: Near corner not brought over properly

- **Visible cue:** the near corner is skipped, it is dropped on the box instead of pressed flat, its tip is left hanging past the far edge instead of folded back, or a box face still shows at the front.
- **SOP rule broken:** Step 2.1, the right gripper lifts the near corner over the box, lays it flat on the top face, presses it down, and folds the tip back if it hangs past the far edge.
- **Coaching note:** pinch the tip, take it straight over, and press it down flat before the far corner moves.

Violation: Far corner wrong

- **Visible cue:** the far corner is laid under the near corner instead of on top of it, its tip is taped, left raw, or not folded under so a point sticks out, it is bunched instead of flat, or a box face still shows at the back.
- **SOP rule broken:** Step 2.2, the right gripper folds the tip of the far corner under, then lays it flat on top of the near corner and presses it down.
- **Coaching note:** far corner always on top, tip folded under, pressed flat. That is what the knot holds down.

Violation: Flaps let go before the knot

- **Visible cue:** the holding gripper opens while both side corners are still out on the table, a flap springs back or slides off the top face, and the wrap carries on without laying it again.
- **SOP rule broken:** Steps 2.2 and 3.1, the left gripper holds both flaps down until the right gripper takes over the hold, and the right gripper holds them until the left gripper's rope is holding them down.
- **Coaching note:** one gripper is always on the flaps until the knot holds them. Hand the hold over, do not just let go.

Violation: Side corner not gathered

- **Visible cue:** a side corner is grabbed at the middle of an edge instead of at its tip, or it is taken up as a wide sheet of cloth with no gathering slide, so the knot comes out wide, flat, and bulky.

- **SOP rule broken:** Steps 3.1 and 3.2, each gripper pinches its side corner at the tip and slides down toward the box until the cloth squeezes into a gathered rope.
- **Coaching note:** pinch the tip and slide down to gather. A rope ties tight; a sheet of cloth does not.

Violation: Knot not square

- **Visible cue:** the second pass is made the same way round as the first, the knot stands up on end rather than lying flat, or it slips loose as soon as the grippers let go and the episode carries on anyway.
- **SOP rule broken:** Step 3.3, the second pass is made the opposite way round to the first, so the knot lies flat on the top face and holds.
- **Coaching note:** first pass one way, second pass the other way. Same way twice slips every time.

Violation: Knot loose or off center

- **Visible cue:** at the end of Step 3 the knot sits off the middle of the top face or over an edge, the cloth is slack and lifts off a face, a flap is loose under the knot, or the two tails are clearly different lengths.
- **SOP rule broken:** Step 3.3, both grippers pull their ropes apart until the square knot is tight and sits on the middle of the top face with the cloth tight on every face.
- **Coaching note:** seat the first half knot in the middle before the second pass, then pull both ropes apart evenly.

Violation: Cloth pulled tight by dragging the bundle

- **Visible cue:** one gripper holds a rope still while the other drags, pushes, or slides the bundle across the table to take up slack, or the bundle skids out of the middle of the working area during the knot.
- **SOP rule broken:** Step 3.3, the cloth is pulled tight by pulling the two ropes apart, never by dragging the bundle.
- **Coaching note:** both grippers pull away from the knot. The bundle stays put.

Violation: Gift card missing, extra, or loose

- **Visible cue:** no card is on the bundle at the end of Step 4, more than one card is taken off the card stack, or the card rests loose on the bundle and falls, slides, or is blown out of place when the right gripper opens.
- **SOP rule broken:** Step 4, the right gripper tucks exactly one card under the knot so it stays put when the gripper opens.
- **Coaching note:** one card, tucked in, and confirm it holds before the gripper opens.

Violation: Card tucked in the wrong place or the wrong way

- **Visible cue:** the card is pushed under a flap, under the cloth, or under the band instead of under the knot, it faces writing side away from the front edge, it lies flat on the top face instead of standing between the tails, or it is forced in so far that the knot loosens.
- **SOP rule broken:** Step 4, the card goes under the near side of the knot, standing between the two tails, writing side toward the front edge, without loosening the knot.
- **Coaching note:** bottom edge under the knot, writing to the front, stop as soon as it stands.

Violation: Band applied wrong

- **Visible cue:** the band is left sitting on the knot or over the tails, it lands high on the top face or low enough to slide off the bottom, it comes out twisted, it hangs slack around the body, more than one band is put on, or the band step is skipped altogether.
- **SOP rule broken:** Step 5, one band goes flat around the middle of the body, clear of the knot, the tails, and the card.
- **Coaching note:** middle of the body, flat and snug, nothing else caught in it.

Violation: Band forced on

- **Visible cue:** the band is rolled or dragged up the outside of the bundle from the table, it is not opened wider than the bundle before it is lowered, it is released before it reaches the middle of the body, or it snaps down and drags the card or a tail out of the knot.
- **SOP rule broken:** Step 5, both grippers open the band wider than the bundle, lower it over the knot, the tails, and the card without touching them, then open together at the middle of the body.
- **Coaching note:** open it wide first and lower it straight down. A band that has to be forced is not open enough.

Violation: Wrong grip on the carry

- **Visible cue:** at Step 6 the gripper closes on the knot, a tail, the card, or the band to lift the bundle, or it grips above the band instead of at the middle of the right side of the body.
- **SOP rule broken:** Step 6, the right gripper closes on the body at the middle of its right side, below the band and clear of the knot, the tails, and the card.
- **Coaching note:** carry it by the body. Lifting by the knot pulls the wrap loose.

Violation: Bundle staged wrong in the lane

- **Visible cue:** at Step 6 the bundle is moved with the left gripper, is left in the working area, is dragged to the lane, lands well off the lane, is set on top of another bundle, is set touching the bundle beside it or out of line with it, or lands knot down or with the card facing away from the front edge.
- **SOP rule broken:** Step 6, the right gripper carries the bundle into the outbound lane and sets it down knot up, card facing the front edge, squared to the lane, beside any bundle already there with a clear gap, and never stacked.

- **Coaching note:** finished bundles stand side by side in one row, knot up. Stacking crushes the knot underneath.

Violation: Work done out of order

- **Visible cue:** a side corner is knotted before both flaps are over the box, the band goes on before the knot is tied, the card is tucked before the knot is tight, or the bundle is carried to the lane before the card or the band is on.
- **SOP rule broken:** Steps 2 to 6, both flaps go over first, then the knot, then the card, then the band, then the carry to the outbound lane.
- **Coaching note:** flaps, knot, card, band, lane. Nothing moves up the order.

Violation: Check skipped

- **Visible cue:** a gripper moves straight on to the next step without the look the step calls for — no look at the four cloth corners after the box lands, no look at the knot after it is pulled tight, no look at the card after the gripper opens, or no look at the band before the bundle is lifted.
- **SOP rule broken:** Steps 1, 3, 4, and 5, each check is made before the next step starts, and a failed check is retried before going on.
- **Coaching note:** look, then move. A check made after the bundle is in the lane is too late to fix anything.

Violation: Gripper opened away from a release point

- **Visible cue:** a gripper opens over the order spot, over the cloth stack, over the band post, over bare table, or in mid-air, and the cloth, box, card, band, or bundle drops or lands where it was not meant to.
- **SOP rule broken:** Steps 1 to 6, a gripper opens only over the working area, the band post, the card stack, or the outbound lane, and only after the thing it holds is settled.
- **Coaching note:** carry it all the way to its spot, let it settle, then open.

Violation: Arms swapped roles

- **Visible cue:** the right gripper lays out the cloth or carries the box from the order spot, the left gripper folds the near or far corner over, takes the right corner, tucks the card, lifts a band off the band post, or carries the bundle to the outbound lane, or one gripper reaches across the table for the other side's corner or supply.
- **SOP rule broken:** Steps 1 to 6, the left gripper owns the cloth, the box, and the left corner, and the right gripper owns the folds, the right corner, the card, the band, and the carry to the lane.
- **Coaching note:** each arm stays on its own side. Nothing crosses the table for a corner or a supply.

Violation: Supply station disturbed

- **Visible cue:** the cloth stack is knocked over or left in a heap, the card stack is pushed out of place or scattered, the band post is struck and shifts or tips, or bands or cards are left lying in the working area at the end of the episode.
- **SOP rule broken:** Steps 1, 4, 5, and 7, take one thing at a time from each station, leave the stations as they were found, and finish with a clear working area.
- **Coaching note:** come in over the top of a stack, take one, and leave the rest square.

Violation: Wrap damaged in handling

- **Visible cue:** the cloth snags, catches on a gripper, or is pulled out of the knot during the card tuck, the banding, or the carry, a tail is pulled through the knot, or the card is bent, creased, or torn, and the episode carries on with the damage.
- **SOP rule broken:** Steps 4 to 6, the card and the band go on without loosening the knot, and the bundle is carried by the body so the wrap stays as it was tied.
- **Coaching note:** ease off the grip near the knot and come in on the side of the body, clear of the cloth ends.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a box, a cloth, a card, a band, or a finished bundle falls to the table or the floor short of its spot, the box is knocked off the cloth, a bundle already in the lane is struck and shifts or tips, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a box face showing, a loose or missing knot, no card or no band, the bundle still in the working area, a cloth, card, or band left loose on the table, a supply station out of place, or an arm away from home.
- **SOP rule broken:** Step 7, confirm the wrap, the knot, the card, the band, the bundle in the lane, the supply stations, and the clear working area; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).

- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective box:** crushed, split, or will not stay closed, so it cannot be wrapped cleanly. Replace it before the next episode.
- **Defective cloth square:** it arrives torn, frayed at a corner, or too small to knot on the top face. Replace it before the next episode.
- **Defective band:** it arrives perished, snaps on its own while being opened, or has stretched slack so it will not hold the wrap. Replace it before the next episode.
- **Defective gift card:** it arrives bent, creased, or torn so it will not stand under the knot. Replace it before the next episode.
- **The cloth stack, the card stack, the band post, or the outbound lane shifts out of place** on its own during the episode, with no arm touching it.

Annotation subtasks (from SOP)

1. Lay the cloth square flat as a diamond in the working area
2. Set one box in the middle of the cloth, top face up
3. Fold the near corner over onto the top face
4. Fold the far corner over on top of it
5. Gather the left corner and bring it to the middle of the top face
6. Gather the right corner and bring it to the middle of the top face
7. Tie the square knot on the top face
8. Tuck one gift card under the knot
9. Open one band and lower it onto the middle of the body
10. Stage the bundle to the outbound lane
11. Confirm the order and end the episode